



DAWlingo: An Educational Tool Teaching Beginners How to Use Digital Audio Workstations

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Motivation

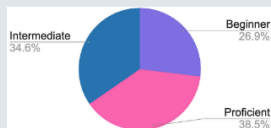
Our team is composed of **five** students with a common interest in music production, but we feel that the process of learning how to use music production software could be made much simpler. We wanted to bridge the gap for beginners who are interested in production, and to empower them to become **DAW-literate** through DAWlingo.



User Research

Interviews

We first conducted **26** semi-structured interviews.



The main struggles that people faced when learning DAWs were **time** and **difficulty**. From this, we created a survey for people with **little to no experience** in music production.

Survey Results

These were the biggest reported barriers preventing people from learning music production:

- **75%** Not enough time to learn
- **66.7%** DAW Softwares are too complicated
- **50%** Takes too long to become proficient

Personas

- People with little to no musical background
- Musicians that have little to no experience producing music

Problem statement

How can we **streamline** the learning process of **DAWs** to make music production **more accessible** to all individuals?

Methodology

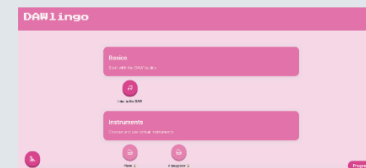
1. Decided on our three main interaction flows: Completing a **guided lesson**, Using the tools in a **freeform playground**, Viewing **progress summary**.
2. Created a paper prototype to map out these interaction flows.
3. Designed one complete lesson to teach the user about basic DAW tools/buttons.
4. Designed playground for user to play with tools that have been learned.
5. Tested this interface with multiple users to find initial issues and bugs.
6. Refined our interface based on initial user tests.
7. Created 2 more lessons, Piano and Arpeggio
8. Performed user evaluations.
9. Noted user behavior and any difficulties encountered iteratively updating the system.



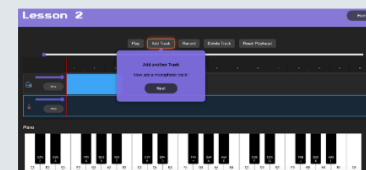
Try it out on desktop!
dawlingo.vercel.app

Implementation

We used **React** to build our system.



We used the **Tone.js** library to implement our DAW features.



Upon lesson completion, users can practice using the unlocked tools in the playground.



A summary serves as a **progress tracker** the user can refer to.

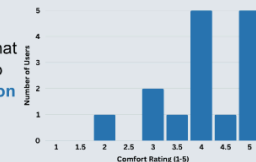
Evaluation

We conducted **15** usability tests, where each potential user completed 2 lessons, Intro to DAW and Piano, and then applied the skills they learned in the Playground Workstation afterwards.

After the user completed the lesson, each user was asked a series of questions and we were able to conclude the following:

The average lesson took **107.87 seconds** and when rating how comfortable they felt applying their new skills in a different DAW system after a lesson, DAWlingo received an average score of **4.07** (scale of 1-5).

Thus, this indicated that DAWlingo was able to address the **2 common concerns** users expressed during our user research phase.



Conclusion/Next Steps

We learned through user testing that our guided lessons are **easy to follow** and helped users feel **more confident** in using DAWs.

Our next steps would be to create lessons for the more complicated/difficult controls and functionality available in DAWs, which is where the true inspiration for our system originally came from.

We also found that users tend to try to click through the lessons without reading and following the instructions, so we would like to control and guide our lessons even more, with that in mind.